

DFA: Formal definitions



A **finite-state machine** or a **deterministic finite-state automaton** (DFA) has five components:

- ▶ the input alphabet Σ ,
 - ▶ a finite set of states Q ,
 - ▶ a transition function $\delta: Q \times \Sigma \rightarrow Q$
 - ▶ a start state $s \in Q$, and
 - ▶ a subset $A \subseteq Q$ of accepting states.
- Handwritten notes:*
- Next to Σ : $\{0, 1\}$
- Next to Q : $\{s, m, a\}$
- Next to δ : $\delta: Q \times \Sigma \rightarrow Q$
- Next to s : s
- Next to A : $\{a\}$

Acceptance

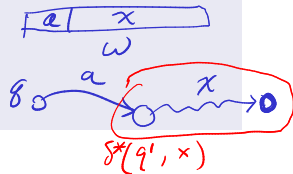
word DFA accepts w

One step move: from state q with input symbol a , the machine changes its state to $\delta(q, a)$.

Extension: from state q with input string w the machine changes its state to $\delta^*(q, w)$ defined as

$\left\{ \begin{array}{l} \varepsilon \\ \text{ax input string} \end{array} \right.$

$$\delta^*(q, w) = \begin{cases} q & \text{if } w = \varepsilon, \\ \delta^*(\delta(q, a), x) & \text{if } w = ax. \end{cases}$$



The signature of δ^* is $Q \times \Sigma^* \rightarrow Q$.

accepting w

For a finite-state machine with starting state s and accepting states A , it accepts string w iff

$$\delta^*(s, w) \in A.$$

from state s
on input string w
state arrives $\delta^*(s, w)$

Language of a DFA

$L(M)$

For a DFA M , let $L(M)$ be the set of all strings that M accepts. More formally, for

$M = (\Sigma, Q, \delta, s, A)$,

$$L(M) = \{ \underbrace{w \in \Sigma^*} \mid \underbrace{\delta^*(s, w) \in A} \}.$$

We refer to $L(M)$ as the language of M .

Automatic languages²

▷ Product construction $L(M_1)$
for DFA M_1, M_2 $L(M_2)$

Definition (for now)

A language L is "automatic" if there is a DFA M such that $L(M) = L$.

↓
DFA M

Lemma 1 ✱ *closure under boolean operations*

If L_1 and L_2 are automatic languages over alphabet Σ , then

- ▶ $L_1 \cap L_2$,
- ▶ $L_1 \cup L_2$,
- ▶ $L_1 \setminus L_2$, and
- ▶ $\Sigma^* \setminus L_1$,

are also automatic.

notes: if M_1, M_2 s.t. $L(M_1) = L_1$
 $L(M_2) = L_2$

suppose s.t. $L_1 = L(M_1)$ then M_1 exists

$L(M)$
 $= L(M_1)$
 $\cap L(M_2)$
↑
 $\cup \cap -$
if M s.t. $L(M) =$

The set of automatic languages is closed under these boolean operations.

²Taken directly from Erikson's lecture notes

Other ways to combine DFAs?

Given two languages L_1 and L_2 , we can combine them in various ways using Boolean operations (i.e., $\cap$, $\cup$, etc.).

What else can we do?

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Given two languages L_1 and L_2 , we can combine them in various ways using Boolean operations (i.e., $\cap$, $\cup$, etc.).

What else can we do?

- Concatenation: $L_1 \cdot L_2$, defined as

$$\{x \cdot y \mid x \in L_1, y \in L_2\}$$

- Kleene closure: L_1^* .

string $w \in L_1^*$ is

① $w = \epsilon$

② $w = x \cdot y$ for all $x \in L_1, y \in L_1^*$

Interesting questions

We know that the set of automatic languages is closed under Boolean operations.

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- ▶ Is it closed under concatenation?
- ▶ Is it closed under taking Kleene closure?

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Spoiler:

Interesting questions

Let $L_1 = L(M_1)$ and $L_2 = L(M_2)$
Given M_1 and M_2 DFA.
 $\Rightarrow$ DFA M s.t. $L(M) = L_1 \cdot L_2$

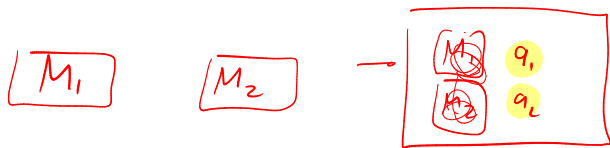
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Questions

- ▶ Is it closed under concatenation? *
- ▶ Is it closed under taking Kleene closure?

* **Spoiler:** Yes, it is (for both operations). We will see the proof, after we learn a required new concept.

Interesting questions



We know that the set of automatic languages is closed under Boolean operations. $Q_1 \times Q_2$

Questions

- ▶ Is it closed under concatenation?
- ▶ Is it closed under taking Kleene closure?

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Closure

Lemma 2

Given two ~~automatic~~^{regular} languages L_1 and L_2 , the following languages are ~~automatic~~^{regular}:

- ▶ $L_1 \cup L_2$,
- ▶ $L_1 \cdot L_2$, and
- ▶ L_1^* .

Closure

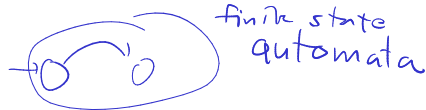
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More over $\emptyset$ and a language containing a single string are also ~~automatic~~.

Closure



More over, $\emptyset$ and a language containing a single string are also automatic.

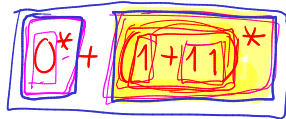
Lemma 2

These are automatic languages

- ▶ The empty set, ✓
- ▶ A language containing one string, ✓
- ▶ $L_1 \cup L_2$ for automatic languages L_1 and L_2 , ✓
- ▶ $L_1 \cdot L_2$ for automatic languages L_1 and L_2 , and
- ▶ L^* for automatic languages L .

Doese this look familiar?

Regular languages



Definition: regular languages

A language L is **regular** if and only if it satisfies one of the following conditions:

- ▶ L is empty;
- ▶ L contains one string (can be the empty string ε);
- ▶ L is a union of two regular languages;
- ▶ L is the concatenation of two regular languages; or
- ▶ L is the Kleene closure of a regular language.

→ "ähnlich" to grammar (pattern)

$\text{regular}(L) \equiv "L \text{ is regular}"$
 $\text{automatic}(L) \equiv "L \text{ is automatic}"$

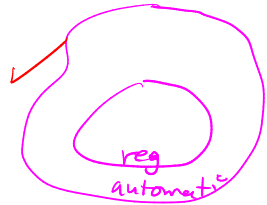
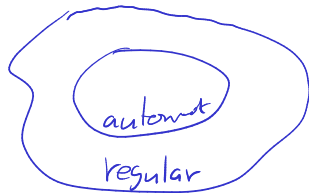
$\Rightarrow$

Every regular language is automatic

$\checkmark \forall \text{ language } L. [\text{regular}(L) \Rightarrow \text{automatic}(L)]$

Big question:

$\forall \text{ lang } L [\text{automatic}(L) \Rightarrow \text{regular}(L)]$



Is every automatic lang regular?

⇒

Every regular language is automatic

Big question:

⇐

Is every automatic language regular?

Spoiler:

Reg 0 $\rightarrow$ ✓ ① $\emptyset$ $\leftarrow$ $\rightarrow \emptyset^{0,1}$
 ✓ ② lang with 1 str
 ✓ ③ $L_1 \cup L_2$ if L_1, L_2 in reg 0.



$\Rightarrow$

Every regular language is automatic



M_3

M_1, M_2

Big question:

$\Leftarrow$

Is every automatic language regular?

*

Spoiler: Yes, it is. We will see some idea on how to prove this.



Every regular language is automatic



Big question:



Is every automatic language regular?

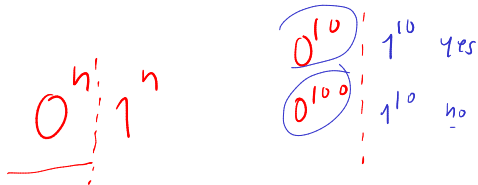
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Theorem 3

A language L is regular if and only if there exists a DFA M such that $L(M) = L$.



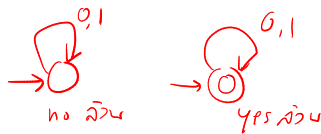
Key idea



If you have finite states, you can't possibly distinguish between strings in the language and strings not in the language.

Basic question

How can you show that you need at least two states?



Basic question

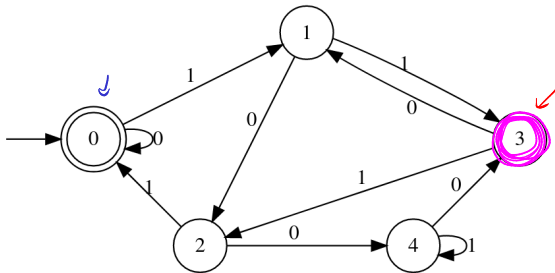
How can you show that you need at least two states?
Let's see how a DFA works.

Another example

$\begin{cases} \epsilon \\ 0 \\ 00 \\ 000 \\ 0000 \end{cases}$

ถ้า $x \in L$ หรือ

$\bullet 11 \bullet x \in L$



$\bullet 11100 \bullet x \notin L$

$\Downarrow$
 $L(M) \neq L$

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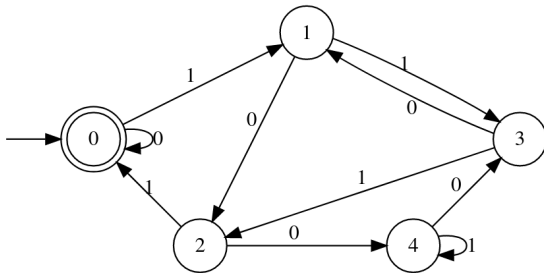
บวก: $\underline{M'}$ ที่ $L(M') = L$
 $(\Sigma, Q, \delta, s, A)$

$\Rightarrow$

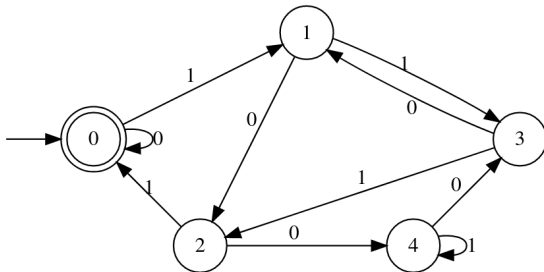
$$\delta^*(s, \underline{11}) \neq \delta^*(s, \underline{11100})$$

Another example

$\widehat{w_1}, \widehat{w_2}, \widehat{w_3}, \widehat{w_4}, \widehat{w_5}$

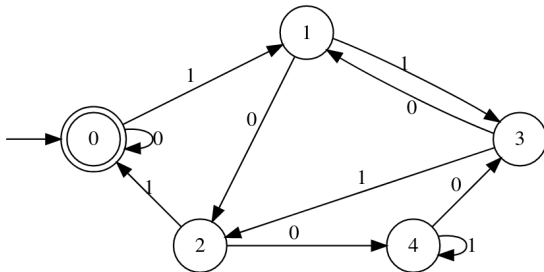


Another example



If string x and y reach the same state in a DFA, for any string z , both xz and yz must reach the same state.

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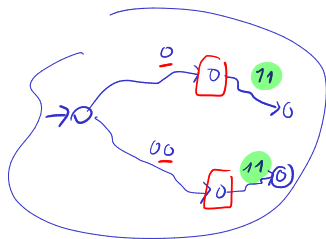
If string x and y reach the same state in a DFA, for any string z , both xz and yz must reach the same state.

In other words, a DFA accepts xz iff it accepts yz .

Distinguishing suffixes

Consider language $L = \{0^n 1^n \mid n \geq 0\}$.

Consider $x = \underline{0}$ and $y = \underline{00}$. Consider suffix $z = 11$.



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We have that

$$xz = 0\mathbf{11} \notin L,$$

but

$$yz = 00\mathbf{11} \in L.$$

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What can you say about a DFA M such that $L(M) = L$?

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What can you say about a DFA M such that $L(M) = L$?

Definition

For strings x and y , string z is a distinguishing suffix with respect to L if exactly one of xz and yz is in L .

state $\in M$ now x
now y
now 11

Distinguishing suffixes

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Definition

For strings x and y , string z is a **distinguishing suffix** with respect to L if exactly one of xz and yz is in L .

w_1
 w_2
 w_3
 w_4
 w_5
 w_6
 $\vdots$
 w_{10}

$\forall w_i, w_j \ (i \neq j)$
 $\exists z$
 $\exists w_i \cdot z \in L$
 $w_j \cdot z \notin L$

Fooling sets

A **fooling set** for a language L is set F of strings such that every pair of strings in F has a distinguishing suffix.

Example: The set $\{0, 00, 000\}$ is a fooling set for $L = \{0^n 1^n \mid n \geq 0\}$.

A large fooling set

Lemma 4

The set $\{0^n \mid n \geq 0\}$ is a fooling set for $L = \{0^n 1^n \mid n \geq 0\}$.

"F

Proof.

WtW: $\forall n, x, y \in F$ $\forall x \neq y$

or distinguishing suffix

$\Downarrow$

or $\exists z$ $\forall$ ① $xz \in L$
 $yz \notin L$

② $xz \notin L$
 $yz \in L$



A large fooling set

Lemma 4

The set $\{0^n \mid n \geq 0\}$ is a fooling set for $L = \{0^n 1^n \mid n \geq 0\}$.

Proof.

Let $F = \{0^n \mid n \geq 0\}$,

q: distinguish z (2) k
PLAN
• $xz \in L$
• $yz \notin L$

พิจารณา $x, y \in F$ ที่ $x \neq y$.

ถ้าเลือก $x = 0^i$ บวก: $y = 0^j$ ที่ $i \neq j$

ให้ $z = 1^i$. สังเกตว่า z จะ

distinguishing suffix ของ x บวก: y

บวก:

$$xz = 0^i 1^i \in L$$

$$\text{บวก: } yz = 0^j 1^i \notin L \text{ เพราะว่า } i \neq j$$

ดังนั้น F เป็น fooling set มากขึ้น



Observation

If language L has an infinite fooling set, L is not regular

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Lemma 5

Language $L = \{0^n 1^n \mid n \geq 0\}$ is not regular.

Proof.

We previously establish that the set $F = \{0^n \mid n \geq 0\}$ is a fooling set for L . Since F has infinite size, from the observation, we know that L is not regular. □

Lemma 6

For $\Sigma = \{0, 1\}$, the language $L = \{\underline{w}w^R \mid w \in \Sigma^*\}$ is not regular.

Proof.

Let $F = \{0^n \underline{1} \mid n \geq 0\}$. Now $x, y \in F$ and $x \neq y$. Let $x = 0^i \underline{1}$, $y = 0^j \underline{1}$ $i \neq j$.

Let $z = 0^i \underline{1}$ (assuming $i < j$).

$xz = 0^i \underline{1} 0^i \underline{1} \in L$

$yz = 0^j \underline{1} 0^i \underline{1} \notin L$ because $i \neq j$.

Thus x & y have distinguishing suffix $\Rightarrow F$ is a fooling set of infinite size.

$\Rightarrow L$ is not regular.

$$\begin{array}{r} 0/0 \\ \hline 0000 \end{array}$$



$L = \{0^{2^n} \mid n \geq 0\}$: Proof 1

Lemma 7

For $\Sigma = \{0\}$, the language $L = \{0^{2^n} \mid n \geq 0\}$ is not regular.

Proof.



$L = \{0^{2^n} \mid n \geq 0\}$: Proof 2

Lemma 8

For $\Sigma = \{0\}$, the language $L = \{0^{2^n} \mid n \geq 0\}$ is not regular.

Proof.



$L = \{0^{2^n} \mid n \geq 0\}$: Proof 3

Lemma 9

For $\Sigma = \{0\}$, the language $L = \{0^{2^n} \mid n \geq 0\}$ is not regular.

Proof.



$L = \{0^p \mid p \text{ is prime}\}$: Proof 1

Lemma 10

For $\Sigma = \{0\}$, the language $L = \{0^p \mid p \text{ is prime}\}$ is not regular.

Proof.



$L = \{0^p \mid p \text{ is prime}\}$: Proof 2

Lemma 11

For $\Sigma = \{0\}$, the language $L = \{0^p \mid p \text{ is prime}\}$ is not regular.

Proof.

